VidiVerify

Quality control for your video files

User Manual

Desktop application for integrity checking of video files

Version 1.4.7

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Basic and PRO editions | Edition for end users and technical professional

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1. Overview and purpose

1.1 What is VidiVerify?

VidiVerify is a professional desktop application for the systematic integrity checking of video files. The software inspects media files for errors, damage and completeness before they are archived, forwarded or used in production. Comparable to a technical inspection for a vehicle, VidiVerify performs this role for digital video files: a structured, reproducible and documentable quality control.

The application is aimed both at private users who occasionally check media files, and at professionals in archiving, broadcasting, post-production, video forensics, education and IT administration who need a reliable tool for daily use.

1.2 Typical problems

Digital video files can suffer damage in many ways without it being immediately obvious during playback. Typical causes of hidden errors include:

- Aborted uploads or interrupted downloads
- Copy errors over unstable networks or defective storage media
- Incomplete exports from video editing or conversion software
- Lost packets during streaming or recording
- Gradual corruption from age-related decay of storage media
- Header structures that no longer match the actual file content

> **[i] Why a dedicated check?**
> A file that opens and plays partially is not automatically free of errors. Defects can hide in the last tenth of the file, individual frames may be damaged, or audio and video streams may drift out of sync. Only a thorough technical check reveals such problems reliably.

1.3 The idea behind the application

VidiVerify brings together a coordinated set of proven, high-performance video plug-ins under a clear, unified user interface. These include FFmpeg and FFprobe for decoding and metadata analysis, MediaInfo for detailed media information, VLC for video preview, Magika for content-based file type detection, and OpenCV for extended image analysis. Each plug-in contributes its own findings, and together they yield insights that exceed what any single tool can offer. The application handles the elaborate parameterization automatically and presents the results in a clear, consistent form.

This makes professional file checking possible without deep knowledge of the underlying command-line tools. The proprietary MAP module (Media Analysis Protocol) consolidates all check results in a user-friendly view with clear color codes.

1.4 A concrete example: what VidiVerify does for you

To illustrate the practical benefit, the following example shows the effort involved in checking a video file without VidiVerify, using the underlying command-line tools directly. A typical, complete check would have to call several plug-ins one after the other and then merge their outputs manually — something like this:

```
ffmpeg -hide_banner -v error -stats -xerror \
       -err_detect +explode+crccheck+bitstream+buffer \
       -fflags +discardcorrupt+genpts \
       -threads 4 -i "sample_video.mkv" \
       -map 0:v? -map 0:a? -map 0:s? \
       -c copy -f null - 2> check_protocol.log

ffprobe -v error -show_format -show_streams \
        -show_packets -show_frames \
        -of json "sample_video.mkv" > metadata.json

mediainfo --Output=XML "sample_video.mkv" > mediainfo.xml
```

Each block of commands has to be typed manually, adapted to every format, fed with the right paths, and finally interpreted. The resulting log files contain thousands of lines of raw output that still need to be evaluated. Error messages appear in technical English, individual findings are not automatically sorted by relevance, and the result is just plain text. In practice, a solid assessment also requires additional plug-ins — for example MediaInfo for deeper container data, Magika for content-based type checking — each with its own invocation, parameters and output format.

VidiVerify performs all these steps automatically: the file is added by drag-and-drop, the check starts at the press of a button, the combined plug-in results appear in MAP with color ratings, and they can be exported directly as a protocol. What took several minutes of manual command-line work becomes a single click.

1.5 Three check levels

The application offers three check levels that build on each other and differ in depth and runtime:

Check level	Runtime	Purpose
Precheck	A few seconds	Verifies reachability, type and metadata of the file automatically on load
Integrity check	30 seconds (sample)	Decodes a defined section of the file and detects typical structural defects
Full check	Minutes to hours	Decodes the entire file and reveals hidden defects as well

2. Main features at a glance

VidiVerify combines a set of features that together enable end-to-end quality control of video files. The following overview lists the most important capabilities of the current version. Features that are only available in the PRO edition are marked accordingly.

2.1 File analysis and metadata

- Automatic reading of container, video and audio streams
- Display of resolution, aspect ratio, codec, bitrate, sample rate and bit depth
- Detection of file type and format family from the header signature (magic bytes)
- Quality classification by standard classes from mobile resolution to Ultra HD

2.2 Check procedures

- Precheck runs automatically when a file is loaded
- Integrity check with a 30-second sample
- Full check across the entire file
- Optional strict mode in which the first decoding error already aborts the check

2.3 Protocol generation and export

- The proprietary MAP (Media Analysis Protocol) with clear color coding
- Export of MAP as text protocol and as PDF protocol
- Dedicated "Diagnostic deep analysis" section in the protocol once the Diagnostic tool has been used
- MAP+ as a high-quality PDF archival document in professional layout PRO
- Export of an NFO file for video library systems such as Kodi, Jellyfin and Emby PRO
- File card with structured media and quality information
- Preview window with zoom and color highlighting

2.4 Robustness and diagnostics

- Diagnostic tool with three-stage deep analysis (streamcopy probe, ffprobe deep scan, decode layer) for difficult cases
- Automatic escalation offer: when the standard check reports anomalies, VidiVerify asks "Start diagnostics?" with a single click
- AI-assisted type integrity via the integrated Google Magika plug-in (Apache-2.0, content-based, independent of extension and header — a permanent part of VidiVerify since v1.4.7)
- Write-protected protocols for exceptions and runtime events
- Separate log files for subprocesses, exceptions, hangs and runtime

- Watchdog for the user interface
- Stable, anonymized device identifier for support packages
- Automatic creation of a support package including relevant system information, optionally with privacy scrub (paths, hostnames, IP/MAC masked)

2.5 Usability

- Refined Overview tab: three check steps as equal-weight cards with pulsing status indicators, plain-text findings and recommendations
- Photo-style preview frame with passe-partout, soft crossfade (slideshow) and automatic detection of cover, snapshot or image series
- Live telemetry directly below the controls: system CPU, check process CPU, RAM, network and read rate
- Honest progress bar with adaptive wall-clock estimation, no more jumps or stalls
- Three user-interface designs: Design mode, Windows classic, PLEX mode
- Multilingual user interface in German, English and Chinese — first-launch language is auto-detected from the system locale
- Three CPU performance modes: Gentle, Balanced, High-Performance
- Drag-and-drop support for video files
- "Protocol" tab in the main window with detailed runtime information about the internal flow

2.6 Updates and maintenance

- Independent update check for the application itself
- Separate update check for the integrated plug-ins
- Built-in download and installation wizard for plug-ins
- Verification of archive integrity via SHA-256 hashes

2.7 Planned additional features (preview)

The following features are in planning and will become available in upcoming versions. A detailed description can be found in Chapter 13.

- Embedded video preview with a statistics window as a modal detail window opened from the "Media Analysis" tab
- "Multi-Analysis" tab for batch processing of multiple video files PRO
- "Media Repair" tab for rule-based repair of damaged files PRO
- "Compatibility" tab with rule-based assessment based on technical media data
- "DVD Check" tab for video and data DVDs as well as Blu-ray structures
- "Statistics" tab with adaptive runtime prediction PRO

- "Perceptual QA" tab for visual quality assessment PRO
- Yutup: YouTube upload forecast PRO
- Porting and refactoring for macOS
- Additional languages: Spanish, Hindi, Russian, Portuguese

3. License model: Basic and PRO

VidiVerify follows a transparent, two-tier license model. The Basic edition is permanently free of charge for private users. Commercial users require a license and automatically gain access to the PRO edition with additional features.

3.1 Basic edition

The Basic edition contains all core features for reliable integrity checking of video files. It can be used without restrictions and is suitable for private use, learning purposes and non-commercial deployments.

- Precheck, Integrity check and Full check
- MAP (Media Analysis Protocol) with color coding
- Export as text protocol and PDF protocol
- All visual designs and languages
- All performance and configuration settings
- Plug-in management and update check

3.2 PRO edition

The PRO edition is aimed at commercial users, organizations, archives and specialized use cases. After successful licensing, all PRO features are unlocked automatically. The PRO edition contains the full feature set of the Basic edition and extends it with powerful additional modules.

> **[PRO] PRO features at a glance**
>
> All PRO features in the manual are marked with a yellow badge. Example:
>
> Media Batch PRO
>
> **Already available with PRO:**
>
> - MAP+ as a high-quality archival PDF
> - NFO export for Kodi, Jellyfin and Emby
>
> Planned: Media Batch, Diagnostic mode, Media Repair, Statistics module, VQA, Yutup.

3.3 Automatic unlock

After successful licensing, the application recognizes the valid license automatically on start-up and activates all PRO features without further manual steps. A reinstallation is not required. License checking happens locally; a permanent internet connection is not necessary for operation.

3.4 Transition between Basic and PRO

Private users can switch to the PRO edition at any time by purchasing a license. All existing settings, protocols and check results are preserved. Likewise, when a license expires, no data is lost — the application simply reverts to the Basic edition.

4. Benefits for users and organizations

VidiVerify is deliberately designed as a tool that delivers compelling added value both for individual desktop use and for organizational deployment. The following sections highlight the most important benefits.

4.1 For end users

[+] Save time and avoid risks
Instead of inspecting video files manually or calling several separate tools, VidiVerify runs all relevant checks in a single workflow. Errors are detected before a damaged file is archived or passed on.

- A clear, guided workflow without the command line
- Meaningful, easy-to-read check protocols with color coding
- Immediately visible status indicators instead of cryptic error messages
- Three interface designs for different preferences
- Configurable CPU load so the system remains usable during a check

4.2 For technical professionals

- Full control over time budget, strict mode and CPU settings
- Detailed event logs in the "Protocol" tab for diagnostics and traceability
- Clean separation between program directory and user data
- Openly accessible log files in plain-text format
- Stable, reusable check logic as a foundation for reproducible results
- Automatic creation of a support package for sending to the support team

4.3 For organizations and archives

[*] Quality assurance as a documented process
Every check can be archived as a PDF. This creates an audit-ready protocol that can be used in quality assurance, intake control or forensic documentation. With the PRO edition, MAP+ is additionally available as a high-quality archival PDF.

- Audit-ready logging of check results
- Consistent check criteria across teams and locations
- Open, traceable storage of settings and protocols
- No cloud dependency: checks happen exclusively on the local machine
- Stable device identifier so the support team can match cases to your device

- NFO export for seamless integration into media library systems **PRO**

4.4 Economic benefits

Aspect	Benefit
Reduced research effort	Faulty files are detected early before time is invested in downstream steps
Avoidance of data loss	Early detection of damaged archives saves later recovery costs
Less manual inspection	Automated check runs replace spot-checking individual files
Consistent check standard	All inspectors use the same criteria and protocol structure
Clear license model	Free Basic edition for private use, reliable PRO edition for professional deployments

5. Installation and setup

5.1 System requirements

Component	Recommendation
Operating system	Windows 10 or Windows 11 (64-bit), macOS planned
Memory	At least 4 GB, 8 GB recommended
Disk space	At least 200 MB for the application, plus additional space for plug-ins
Processor	64-bit processor with at least two cores
Screen resolution	At least 1105 x 753 pixels
Permissions	Standard user rights are sufficient; administrator rights are only needed for the initial installation

5.2 Installation steps

1. Download and start the installer.
2. Confirm the target folder (typically C:\Program Files\VidiVerify).
3. Run the installation and start the application.
4. On first launch, the application automatically performs a plug-in check.
5. If plug-ins are missing, a guided download wizard is offered.

> **[+] Simple and safe**
> The setup wizard guides you through the entire process. The required plug-ins can be downloaded either right away or later from within the application.

5.3 First launch and plug-in check

On first launch, a splash screen appears and remains visible for a short time. During this period the application checks whether all required plug-ins are available. If FFmpeg or FFprobe are missing, an installation dialog opens automatically. You can either start the download immediately or skip it; in the latter case some check functions will only become available once the plug-ins have been installed.

5.4 Language selection

The application currently supports three languages and offers them through the Settings dialog.

- German (default on German-language systems)
- English

- Chinese (中文)

Spanish, Hindi, Russian and Portuguese are also planned.

5.5 License activation (PRO)

Commercial users activate the PRO edition through the license module in the Settings dialog. After entering the license key, the PRO features are unlocked automatically. Activation happens locally and remains valid even without an internet connection.

5.6 Uninstallation

Uninstallation is performed via the bundled uninstaller (unins000.exe) in the program directory or through the Windows Control Panel. User data in the user folder is preserved by default and can be deleted manually if desired.

6. Usage, operation and workflows

6.1 Splash screen and main window

After start-up, the main window of the application opens. It is divided into several areas that mirror the typical workflow from file selection to the check protocol.

- File selection with a Browse button and a drag-and-drop area
- Overview tab as the entry point: three check steps as equal-weight cards with pulsing status indicators, plain-text findings and recommended actions
- Photo-style preview frame with a light passe-partout — cover, snapshot or image series are detected automatically
- Information panel with container, duration, video and audio properties
- Check level selection between Integrity check and Full check
- Live telemetry below the controls: system CPU, check process CPU, RAM, network and read rate
- Progress bar with elapsed time and status (adaptive wall-clock estimation)
- Result area with a classified check result (MAP)
- "Protocol" tab with detailed runtime information about the internal flow

6.2 Typical workflow

6. Start the application.
7. Select a video file or drag and drop it.
8. Wait for the automatic precheck (a few seconds).
9. Select the check level: Integrity check or Full check.
10. Start the check.
11. Observe the progress or cancel if needed.
12. Review the result in the MAP (color-coded view).
13. If needed, inspect the details in the "Protocol" tab.
14. In case of anomalies, run the Diagnostic tool (see 6.9).
15. Save the protocol as a text file, MAP PDF or (PRO) MAP+.

6.3 Precheck

The precheck runs automatically as soon as a file is selected. It verifies basic reachability and reads the metadata using the FFprobe plug-in. The result appears immediately in the information panel. This stage delivers an initial assessment within a few seconds without fully decoding the file.

> **[i] Note on network paths**
> Files on network paths may take longer for the precheck depending on connection quality. The Network source policy setting lets you decide whether such sources should be ignored, warned about or blocked.

6.4 Integrity check

The integrity check decodes the first 30 seconds of the file. This section typically contains the most important header structures and index information of the container. Errors in this area indicate structural defects and are detected reliably. The runtime stays manageable because only a defined excerpt is processed.

6.5 Full check

The full check decodes the entire file completely. Only this approach reveals errors that lie in later sections of the file or affect only individual frames. The runtime depends on file size, codec and available processing power, and can reach several hours for long recordings.

6.6 Progress and cancellation

During a check the progress bar shows the current state as a percentage and the elapsed time. A running check can be ended at any time using the Cancel button. The application asks for confirmation to prevent accidental cancellations.

6.7 The "Protocol" tab

In addition to the check result, the main window also offers the "Protocol" tab. It shows detailed runtime information about the internal flow: started processes, timestamps, plug-in messages, detected stream information and technical notes. The tab is especially helpful for technical professionals who want to follow the exact course of a check or to put anomalies into context.

6.8 Check protocol and file card

Once a check is finished, VidiVerify automatically generates the MAP (Media Analysis Protocol). It can be saved as a text file or as a PDF document. If the Diagnostic tool was used, an additional "Diagnostic deep analysis" section appears with an overall assessment and finding. In the PRO edition, MAP+ is additionally available — a high-quality archival PDF for professional filing.

6.9 Diagnostic tool

The Diagnostic tool is the in-depth analysis that delivers answers for stubborn cases. It comes into play when the standard check reports an "Anomaly" and you need a clear cause. Three specialized check layers work together:

16. Streamcopy probe: checks whether the container can be re-wrapped cleanly (a fast structural sanity test) and uncovers timestamp and packet-order problems.
17. ffprobe deep scan: in-depth reading of packet and frame information including timestamp, index and stream structures.
18. Decode layer: experimental decoding of targeted frame ranges, uncovers codec-specific defects that are not visible at the container level.

From these three layers, VidiVerify derives a clear diagnosis — with cause, severity and a concrete recommended action. It reliably identifies tape errors from VHS digitizations, defective DVD grabs, aborted downloads, manipulated streams and damaged containers.

> **[*] Understands quirks instead of false alarms**
> The Diagnostic tool recognizes DVD cell transitions and H.264/HEVC B-frame structures for what they are: normal encoder behavior, not a defect. No more confusing red lights on flawless files.

Automatic escalation offer

When the standard check reports something unusual, VidiVerify asks directly: "Start diagnostics?". One click — and the tool delivers. The check protocol then also includes the "Diagnostic deep analysis" section with an overall assessment and finding.

7. The proprietary MAP (Media Analysis Protocol)

7.1 Idea and goal

The MAP (Media Analysis Protocol) is the proprietary heart of VidiVerify. It translates the complex technical results from in-depth technical checks into a clear presentation that is easy for every user to understand. The central strength of the MAP lies in assessing complicated technical values from in-depth check runs and making them visible through simple color codes.

> **[*] Wow effect: technology meets clarity**
> Raw data from FFprobe, FFmpeg and MediaInfo often consists of thousands of values and parameters. The MAP distills this into statements visible at a glance: color codes instead of columns of numbers, understandable classifications instead of raw technical values. This makes professional video analysis usable for everyone.

7.2 What the MAP delivers

- Summary of all check levels in a single overview
- Color-coded display of the overall result (see Appendix B)
- Key technical values classified in understandable terms
- Notes on anomalies in plain language
- Link between the raw protocol and the user-friendly assessment
- Direct export as text or PDF

7.3 Structure of the MAP

The MAP is divided into several sections that build on each other and together provide a complete picture of the checked file:

Section	Contents
Header	File name, path, file size, check timestamp
Basic technical data	Container, streams, resolution, codec, duration
Audio	Format, sample rate, bit depth, channels, bitrate
Quality classification	Categorization into SD, HD, Full HD, 2K to 4K, above 4K
Check result	Color-coded status display (OK, Warning, Error)
Notes	Classification notes and anomalies
Runtimes	Time spent on precheck, integrity check and full check

7.4 MAP export

The MAP can be exported directly from the application. Two formats are available:

- Text protocol (.txt) — for quick technical filing and further processing
- PDF protocol (.pdf) — for archiving, sending and documentation

[PRO] MAP+ as an archival PDF (PRO)

The PRO edition additionally offers MAP+: a high-quality PDF archival document with a professional layout. It contains an extended presentation, additional technical sections and is optimized for long-term archiving, forensic documentation and check protocols for client handovers.

7.5 NFO export for media library systems (PRO)

In addition to the MAP export, the PRO edition can generate an .nfo file on request. This format is read by video library systems such as Kodi, Jellyfin and Emby and automatically adds technical information to the media collection. The NFO file is created directly from the existing analysis results and conforms to common standards for a robust, long-term maintainable integration.

8. The integrated plug-ins

VidiVerify relies on a set of established, high-performance and specialized plug-ins, each of which handles a clearly defined task. The application coordinates this plug-in ensemble, parameterizes the tools automatically and merges their results in the MAP. For you as the user this creates a single unified interface, while behind the scenes proven tools are at work whose combined insight goes far beyond what any individual plug-in could provide.

8.1 FFmpeg

FFmpeg is a globally recognized plug-in for processing and decoding media files. In VidiVerify, FFmpeg handles the actual decoding during the integrity check and the full check. The application calls FFmpeg with finely tuned parameters so that errors are reliably detected and logged.

- Core task: decoding and checking all streams
- Strengths: broad format support, very robust error detection
- Used in: integrity check and full check

8.2 FFprobe

FFprobe is the sister plug-in to FFmpeg and is used to read metadata and stream information. VidiVerify uses FFprobe in the precheck and to populate the MAP with basic technical data.

- Core task: reading containers, streams and codec information
- Strengths: fast analysis without full decoding
- Used in: precheck and metadata enrichment of the MAP

8.3 MediaInfo

MediaInfo is a specialized plug-in for the detailed analysis of media files. It provides especially extensive and precise information on containers, codecs, streams and advanced parameters. VidiVerify uses MediaInfo as a complementary source and for cross-checking results. Since v1.4.7, MediaInfo also recognizes Dolby Atmos and DTS:X structures in main audio tracks. The main-track selection additionally takes locale and the default tag into account — a German app will find the German audio track even if the container set the default flag elsewhere.

- Core task: deeper metadata analysis with a wide parameter range
- Strengths: very detailed technical information, complementary perspective, Atmos and DTS:X detection
- Used in: complementary analysis, foundation for the compatibility assessment

8.4 VLC

VLC is a well-known plug-in for media playback with the broadest format compatibility. VidiVerify uses VLC as the playback engine for the embedded video preview in the upcoming "Media Test" tab. This allows video files to be played back for testing purposes directly within the application.

- Core task: embedded video playback
- Strengths: broad format support, mature playback
- Used in: preview and visual inspection in the "Media Test" tab

8.5 Magika

Magika is a plug-in published by Google for content-based file type detection. According to Google, it is used productively in their own security and file pipelines. Magika has been a permanent part of the application since v1.4.7. While the classic magic-byte check reads the file header, Magika works model-based on the file content — both methods are shown side by side in the Overview tab. What one method misses, the other often catches, so the two together give a reliable verdict. Magika is licensed under Apache-2.0 and weighs in at around 10 MB.

For swapped or disguised files (for example, a PDF with the extension .mp4 or an EXE inside a video container), VidiVerify clearly marks the result as "not media" with a specific note on how to correct it.

- Core task: content-based file type detection
- Strengths: robust detection even for renamed or manipulated files
- Used in: second opinion to the classic magic-byte check, visible in the Overview tab

8.6 OpenCV

OpenCV is a proven plug-in for image processing and computer-vision tasks. In VidiVerify, OpenCV forms the foundation for advanced image and frame analyses, especially for the planned perceptual quality analysis (VQA).

- Core task: image and frame analysis
- Strengths: extensive image processing algorithms
- Used in: perceptual quality analysis, advanced image evaluation

8.7 Interplay of the plug-ins

Each plug-in handles its specialized task and is orchestrated by VidiVerify. The results flow centrally into the MAP, where they are prepared for the user and color-rated. This architecture creates a reliable, modular and future-proof platform.

9. Configuration and parameters

The application settings are managed in the Settings dialog. It is opened from the menu or via the Settings button in the main window. All changes are saved in the file vidiverify_settings.json in the user folder.

9.1 Paths to the plug-ins

Setting	Description	Default
FFmpeg path	Path to the FFmpeg executable	tools/ffmpeg/ffmpeg.exe
FFprobe path	Path to the FFprobe executable	tools/ffmpeg/ffprobe.exe
MediaInfo path	Path to the MediaInfo plug-in	tools/mediainfo/
VLC path	Path to the VLC plug-in for embedded playback	tools/vlc/
Magika path	Path to the Magika plug-in for DeepCheck	tools/magika/
OpenCV path	Path to the OpenCV plug-in for image analyses	tools/opencv/

9.2 Check behavior

Setting	Description	Default
Strict mode	Even the first decoding error causes the check to abort	On
Time budget	Time limit per check in seconds, 0 means no limit	0 (unlimited)
Maximum protocol lines	Maximum number of lines in the runtime protocol	10,000

9.3 Performance settings

CPU usage can be adjusted via three predefined performance modes that offer a sensible balance between check speed and system responsiveness.

Performance mode	Processor cores	Priority	Typical use
Gentle	One core	Low	Background check during regular work
Balanced	Half of the cores	Normal	Default setting for most cases
High-Performance	All cores	High	Fastest possible check of individual files

9.4 Display and language

Setting	Options
Display language	German, English, 中文 (more planned)

Setting	Options
Interface design	Design mode, Windows classic, PLEX mode
Network source policy	Ignore, Warn, Block

9.5 Update behavior

- Update check for VidiVerify at start-up (can be enabled)
- Update check for the plug-ins at start-up (can be enabled)
- Manual update check from the settings is possible at any time

9.6 Support package

VidiVerify offers automatic compilation of a support package. It contains all relevant system information, protocols and diagnostic data and can be sent to support if needed. The package is compiled locally; sending it is up to you.

- Contains the current version, plug-in versions and runtime protocols
- Contains the anonymized device identifier so support can match the package to your device unambiguously
- Contains no personal data or media content

[+] Principle: safe defaults

The application ships with conservative default settings that are suitable for most use cases. Changes are optional and can be undone at any time.

10. File structure and storage concept

VidiVerify deliberately keeps the program installation separate from user-specific data. The application itself is installed in the program directory, while settings, protocols, plug-ins and optional statistics are stored in a clearly defined local user folder. This keeps the installation tidy, makes runtime data transparent and easy to trace, and allows updates or uninstallations to run cleanly.

10.1 General description

Program folder

```
\VidiVerify
├── VidiVerify  (main program)
└── VidiVerify Uninstall
```

User folder

```
\VidiVerify
├── settings and sources
├── protocols and diagnostics
├── plug-ins and additional modules
└── optional statistics data
```

10.2 Example structure with file names

Program folder

```
C:\Program Files\VidiVerify
├── VidiVerify.exe
├── unins000.exe
└── unins000.dat
```

User folder

```
%LocalAppData%\VidiVerify
├── vidiverify_settings.json
├── module_sources.json
├── logs
│   ├── hangs
│   ├── exceptions.log
│   ├── faulthandler.log
│   ├── runtime.log
│   └── subprocesses.log
├── tools
│   ├── ffmpeg
│   ├── vlc
│   ├── mediainfo
│   ├── magika
│   └── opencv
└── statistics
    ├── performance_history.jsonl
    └── performance_meta.json
```

10.3 Explanation of the most important files

File or folder	Purpose
vidiverify_settings.json	Persistent user settings (language, paths, performance mode)
module_sources.json	Configurable download sources for the plug-ins
logs/runtime.log	Chronological events from regular operation
logs/exceptions.log	Exception errors with full stack trace
logs/subprocesses.log	Protocol of started plug-in processes (FFmpeg, FFprobe, MediaInfo)
logs/faulthandler.log	Information captured during serious crashes
logs/hangs	Diagnostic dumps for unresponsive processes
tools/ffmpeg	Locally installed FFmpeg and FFprobe binaries
tools/vlc	VLC plug-in for embedded playback
tools/mediainfo	MediaInfo plug-in for in-depth metadata analysis
tools/magika	Magika plug-in for the Magika deep check

File or folder	Purpose
tools/opencv	OpenCV plug-in for image analysis
statistics	Optional runtime and performance history

[*] Transparency as a principle

All settings and protocols are stored in open plain-text formats (JSON, text). This means they remain readable outside the application and can be evaluated or archived in a targeted way whenever needed.

This storage concept improves maintainability, avoids unnecessary mixing of program files and user data, and produces a clearly traceable structure. Important settings and runtime data remain file-based, transparent and easy to locate when required. The application can be uninstalled cleanly via the bundled uninstaller.

11. Architecture and technical details

11.1 Overview

VidiVerify is written in Python and uses the bundled standard library together with a selection of well-established additional libraries. The graphical user interface is built on Tkinter, the standard UI toolkit shipped with Python. The actual media analysis is performed by a coordinated set of established, specialized video plug-ins (FFmpeg, FFprobe, MediaInfo, VLC, Magika, OpenCV).

11.2 Component overview

Component	Role
User interface (Tkinter)	Input, display, operation and presentation of results
Check module	Orchestrates Precheck, Integrity check and Full check
MAP (Media Analysis Protocol)	Proprietary evaluation and color coding of the results
Plug-in manager	Handles download, update and launch of the plug-ins
Subprocess manager	Starts and monitors the plug-in processes
Protocol system	Writes the different levels of protocols to files in a structured form
Settings store	Persistence of user settings in JSON format
Watchdog	Monitors the user interface for unresponsive states
License module	Checks and unlocks the PRO edition
Support package module	Automatic compilation of diagnostic information

11.3 Plug-in dependencies

- FFmpeg: decoding and checking of media files
- FFprobe: reading metadata and stream information
- MediaInfo: in-depth media analysis and supplementary metadata
- VLC: embedded video preview and visual inspection
- Magika: content-based file type detection for the Magika deep check
- OpenCV: image processing and frame analysis
- psutil: detailed CPU and process information
- Pillow: image processing for the user interface
- tkinterdnd2: drag-and-drop support
- fpdf: creation of the PDF protocols

11.4 Concurrency

The application uses several concurrency layers to stay responsive. The user interface runs in the main thread, while checks are executed in separate worker threads. Communication between threads happens through a central message queue, which ensures that user-interface updates remain consistent.

11.5 Error handling

Errors are handled and logged at four levels:

Level	Handling
Operational notice	Informational message in the status bar and in the protocol window
Warning	Visible highlight in the protocol, processing can continue
Error	Check is aborted, detailed entry written to the protocol
Exception error	Full stack trace written to exceptions.log, diagnostic dump on demand

11.6 Single-instance mechanism

The application prevents multiple instances from running at the same time through a system-wide lock (mutex) and an additional lock file. When a second instance is launched, a notice informs the user that the application is already running. Stale lock files are cleaned up automatically if the original process no longer exists.

11.7 Stable device identifier

For support purposes, the application generates a stable, anonymized device identifier (Machine ID). It starts with the prefix VV- followed by five characters from a non-confusable character set (no easily mistaken characters). The identifier is derived from a system-wide hardware identification and a built-in salt value. It cannot be used to identify individuals or specific hardware.

11.8 Updates and integrity

Both the application itself and the plug-ins are obtained from signed or hash-secured sources. When a plug-in is downloaded, the SHA-256 checksum of the archive is verified before installation. This reliably rejects tampered or damaged packages.

> **[*] Openness and verifiability**
> All formats, storage paths and protocol structures are designed to remain readable and analyzable even without the application. This keeps full control in the hands of the user.

12. Examples and use cases

12.1 Use case: intake control in a media archive

A media archive receives new video files from production teams and external suppliers every day. Each file is to be checked for structural integrity before it is added to the archive. The full check usually runs overnight so that system load is not affected during working hours.

Typical settings: Full check, High-Performance mode, strict mode enabled, unlimited time budget, export as MAP+ (PRO) for audit-ready storage.

12.2 Use case: quick check before handover

A video editor wants to check a finished episode briefly before passing it on to the client. A full inspection is not needed, but typical container errors must be ruled out. The Integrity check delivers a reliable interim result within a few minutes.

Typical settings: Integrity check, Balanced performance mode, text protocol for internal storage.

12.3 Use case: spot check in an archive audit

An archive employee periodically inspects samples from the holdings to spot gradual media degradation. Each checked file receives a MAP+ PDF protocol that is attached to the entry in the inventory record.

Typical settings: Full check, Gentle performance mode during working hours, MAP+ as archival document (PRO).

12.4 Use case: forensic review

A specialist review in a forensic context calls for traceable checks with complete documentation. VidiVerify records every check with runtime, timestamp, result and classification. The "Protocol" tab and the subprocess log are additionally available as a text-based chain of evidence.

Typical settings: Full check, strict mode enabled, maximum protocol limit, MAP+ as archival document, text protocol in addition.

12.5 Use case: media management with Kodi or Jellyfin

A private user or small operator wants to manage their collection with Kodi or Jellyfin. With the PRO edition, VidiVerify automatically generates an .nfo file that these systems read directly. The media library thus receives complete, technically correct metadata without any manual upkeep.

Typical settings: Integrity check, NFO export enabled (PRO), automatic storage in the media folder.

12.6 Use case: routine IT maintenance

An IT department uses VidiVerify to regularly check media collections stored on file servers. The goal is the early detection of storage problems before whole backup rotations are affected. The update check for the plug-ins is kept active so that up-to-date decoders are always available.

Typical settings: Balanced performance mode, update checks enabled, Media Batch (PRO) for serial checking of entire directories.

13. Planned extensions and outlook

VidiVerify is continuously being developed further. The following features are in preparation and form part of the strategic roadmap. Some of them will belong to the Basic edition, others are PRO features. Final availability, scope and delivery date may change as development progresses.

13.1 NFO export for media library systems

In the current version, VidiVerify already exports the proprietary MAP. In addition, an .nfo file is planned for video library systems.

- Integration with systems such as Kodi, Jellyfin and Emby
- Use of the existing technical analysis information
- Standards-compliant, robust and long-term maintainable implementation

13.2 MAP+ as archival PDF PRO

MAP+ is the high-quality PDF variant of MAP. It is designed as a professional archival document and offers an extended layout with additional technical sections, a more elaborate design and a structure optimized for long-term archival use. MAP+ is aimed at organizations, archives and professional users who pass check protocols on to clients or retain them long-term.

13.3 "Media Test" tab

The "Media Test" tab enables a visual hands-on check with graphical video inspection. The goal is to play the loaded video file directly inside the tool for testing purposes. VLC serves as the playback engine and embeds the video directly into a window of the application.

- Embedded video preview via the VLC plug-in
- Controls: play, stop, forward, backward
- Left pane: video preview window
- Right pane: statistics with frame, time, remaining time and other values
- Visualization of a waveform of the audio stream

13.4 Magika deep check (available since v1.4.7)

The Magika deep check, originally planned for a later version, has been a fixed part of VidiVerify since v1.4.7. It extends the existing check levels with a content-based deep inspection of type integrity, powered by the Magika plug-in. The Overview tab shows the classic magic-byte check and Magika side by side — see Chapter 8.5 for details.

- Robust detection of whether the actual file content matches the expected file type
- Consistency check between file extension, header signature and container structure

- Additional confidence for suspicious, renamed or tampered files

13.5 Batch processing: "Media Batch" tab PRO

The "Media Batch" tab enables batch processing of multiple video files in a single run. A running result analysis is shown during processing.

- Processing of entire directories or file lists
- Running result analysis during the check
- Automatic saving of the MAP protocols per file
- Summary batch protocol with an overview

13.6 Diagnostic tool (available since v1.4.7)

Originally planned as a PRO Diagnostic mode, this feature has been part of the Basic edition free of charge since v1.4.7 as the Diagnostic tool. In Standard mode the checks run with the standard effort, which is sufficient for most cases. If the standard check reports anomalies, deeper plug-in options can be enabled in the Diagnostic tool to determine an unambiguous cause.

The FFprobe plug-in additionally outputs packet or frame information in the Diagnostic tool, for example via the options -show_packets and -show_frames. This enables deeper analysis of problem files and reveals anomalies in timestamps, irregular packet sequences, missing values or unusual frame structures. A streamcopy test also runs to check container handling and stream mapping:

```
ffmpeg -v error -xerror -i "file.mp4" \
       -map 0 -c copy -f matroska NUL
```

This test exposes anomalies in container structure, timestamps or problematic stream configurations. A complete description of the three diagnostic levels and how they interact is given in Chapter 6.9.

13.7 "Media Repair" tab PRO

The "Media Repair" tab offers rule-based repair options for damaged video files. If the check finds a faulty file, VidiVerify proposes a suitable approach based on the specific error message and the repair capabilities of FFmpeg.

- Formatted error report for working through errors in detail
- Rule-based classification of errors with a probability rating
- Concrete repair proposal with an explanation of the procedure
- Repair always operates on a copy, never on the original file
- Verification of the repair result via the "Media Test" tab
- Storage of the result at the user's discretion

13.8 Compatibility assessment

The compatibility assessment provides a rule-based evaluation derived from technical media data. It is not based on FFprobe alone but additionally draws on the capabilities of MediaInfo. The core benefit is that VidiVerify does not make any untenable claim such as "plays everywhere" but instead derives a technically grounded probability rating from objective file characteristics.

- Rule-based evaluation from the combination of FFprobe and MediaInfo data
- Probability statements about playback on typical platforms
- Clear marking of critical characteristics (codec profiles, bit depth, color spaces)
- Substantive, market-ready added value without blanket guarantees

13.9 Statistics module PRO

The statistics module builds an internal, self-learning statistics system inside VidiVerify. It systematically captures real check data and, over time, enables individual runtime predictions.

- Systematic capture of real check data
- Long-term, individual runtime predictions per system
- Detection of differences between systems, CPU modes and storage sources
- Sound data foundation for later analysis and UI features

13.10 Perceptual quality analysis (VQA) PRO

Perceptual quality analysis (VQA) extends the existing analysis functions with an additional evaluation layer that considers the visual perception of a video from a human viewpoint. While MAP and the Yutup prognosis primarily analyze technical parameters and platform behavior, the VQA component answers the question: how good does the video actually look, independent of technical metrics?

- Visual quality assessment from the viewer's perspective
- Complement to technical metrics and platform evaluations
- Use of the OpenCV plug-in for image analysis

13.11 Yutup: YouTube upload forecast PRO

Yutup is an integrated analysis module that evaluates video files technically and visually before they are uploaded to YouTube. This is deliberately not a pure format check; it is a realistic forecast of how the video will look after YouTube re-encodes it.

- Technical assessment of upload suitability
- Quality prognosis after YouTube's re-encoding
- Recommendations for optimization before the upload

13.12 Porting to macOS

The application is to be ported to macOS as part of a refactoring effort. The goal is a consistent user interface and comparable feature set across the supported operating systems.

13.13 Additional languages

Further languages for the user interface are planned:

- Spanish
- Hindi
- Russian
- Portuguese

13.14 License module

The license module manages the activation of the PRO edition, verifies the validity of licenses and unlocks the corresponding features. It works entirely locally and does not require a permanent internet connection.

[*] Outlook

The extensions described here turn VidiVerify into a comprehensive platform for the quality control, analysis and management of video files. The modular architecture allows these features to be introduced step by step without compromising the stability and simplicity of the existing application.

Appendix A: Glossary and terminology

The following overview explains key technical terms from the areas of video files, codecs, audio properties and the integrated plug-ins, as far as they are relevant to VidiVerify. Where the German application interface uses a different label than the natural English term, the English term is given first and the German app label is referenced in parentheses. Entries are sorted alphabetically by the English term.

Term	Explanation
AAC	Advanced Audio Coding, a lossy audio codec, widely used in MP4 and AAC streams
AC3	Audio codec by Dolby, common on DVD and in broadcasting
Atmos	Object-based 3D audio format by Dolby, recognized by MediaInfo since v1.4.7
AV1	Modern, royalty-free video codec with high efficiency, successor to VP9
AVC	Advanced Video Coding, identical to H.264
Bit depth	Number of bits per audio sample or color channel, typically 16 or 24 bits for audio; higher values mean finer resolution
Bitrate	Amount of data per unit of time, typically in kilobits or megabits per second, often an indicator of the quality of a video or audio stream
Blu-ray	Optical storage medium for high-resolution video material, typically using MPEG transport streams
Broadcast	The field of radio and television distribution, including specific container formats such as MXF and transport streams
Chroma subsampling	Technique for reducing color resolution, common variants are 4:4:4, 4:2:2 and 4:2:0
Classification	Assignment of a check result to a defined category, see Appendix B
Codec	Set of rules for encoding and decoding video or audio data, for example H.264, H.265, AAC
Color space	Mathematical description of representable colors; common spaces are BT.709 (HD) and BT.2020 (UHD)
Compatibility assessment (German: Kompatibilitätsbewertung)	Rule-based estimate of playability based on technical media data
Container	File format that bundles video, audio and metadata streams in a single file, for example MP4, MKV, AVI
CRC	Checksum (Cyclic Redundancy Check), used to detect data errors
Magika deep check	Deep type-integrity check using the Magika plug-in; an integral part of the application since v1.4.7

Term	Explanation
Decoding	Converting an encoded media file into displayable images and sound
Device identifier (Machine ID)	Stable, anonymized identifier for the workstation, used by support to recognize a device across submissions
Diagnostic tool (German: Diagnose-Werkzeug)	Three-stage deep analysis (streamcopy probe, ffprobe deep scan, decode layer) for stubborn cases, available since v1.4.7
DTS	Digital Theater Systems, an audio codec, often found on Blu-ray and DVD
DTS:X	Object-based 3D audio format by DTS, recognized by MediaInfo since v1.4.7
EBML	Extensible Binary Meta Language, the structural foundation of the Matroska container format
Emby	Media library system that reads NFO files for metadata
FFmpeg	Plug-in for processing and decoding media files
FFprobe	Plug-in from the FFmpeg package for reading metadata
FLAC	Lossless audio codec (Free Lossless Audio Codec)
Frame	A single image of a video stream
Frame rate	Number of frames per second, typical values: 24, 25, 30, 50, 60 fps
Full check (German: Vollprüfung)	Complete decoding of the entire file for a deep integrity check
H.264	Widely used video codec, also called AVC, standard for HD video content
H.265	Successor to H.264, also called HEVC, more efficient at high resolutions
Hash value	Fixed-length character string calculated from a file for integrity checking; here SHA-256
HDR	High Dynamic Range, extended dynamic range for bright and dark image areas
HEVC	High Efficiency Video Coding, identical to H.265
Integrity check (German: Integritätsprüfung)	Check of the file structure and decodability, here based on a 30-second sample
ISO BMFF	Base Media File Format by ISO, foundation for MP4 and MOV
Jellyfin	Free media library system that reads NFO files for metadata
JSON	Text-based data format for structured storage of settings and metadata
Kodi	Widely used media library system that reads NFO files for metadata
Live telemetry (German: Live-Telemetrie)	Display of system CPU, check-process CPU, RAM, network and read rate directly below the controls, available since v1.4.7
Log (Protocol)	Time-ordered record of events during the operation of the application

Term	Explanation
Magic bytes	Characteristic byte sequence at the beginning of a file that identifies the file format
Magika	Plug-in by Google for content-based file-type detection
MAP	Media Analysis Protocol, the proprietary VidiVerify module for user-friendly presentation of results
MAP protocol	Standard export format of the MAP as a text file or PDF, accessed via the "Save protocol" button in the main window
MAP+	High-quality PDF variant of the MAP as an archival document, a PRO feature
Matroska	Open, flexible container, also known as MKV
Media Batch (German app label: Media Stapel)	Planned tab for batch processing, a PRO feature
Media Repair	Planned tab for rule-based repair of damaged files, a PRO feature
Media Test	Planned tab for visual user inspection with an embedded video preview
MediaInfo	Plug-in for in-depth analysis of media files and stream parameters; since v1.4.7 it also detects Atmos and DTS:X in the main audio tracks
Metadata	Descriptive information about a file, such as resolution, duration or codec
MKV	File extension for Matroska video containers
MOV	Container format by Apple, technically related to MP4
MP3	Lossy audio codec, very widely used
MP4	Widely used container format based on ISO BMFF
MPEG-2	Video compression standard, foundation of DVD and digital television
MPEG-TS	MPEG transport stream, used in broadcasting and on Blu-ray
MXF	Material Exchange Format, professional container format for broadcast and production
NFO	Text-based metadata format for media library systems such as Kodi, Jellyfin and Emby
OpenCV	Plug-in for image processing and computer-vision tasks
Opus	Modern, lossy audio codec with high quality
PCM	Pulse Code Modulation, uncompressed audio data
Perceptual quality analysis	Assessment of the visual perception of a video, see VQA
Performance mode (German: Leistungsmodus)	Predefined profile for controlling CPU load during a check

Term	Explanation
Precheck (German: Vorprüfung)	Short, automatic preliminary check performed when a file is loaded
PRO	Paid license edition for commercial users, with an extended feature set
RIFF	Resource Interchange File Format, the structural foundation of the AVI container
Sample rate	Number of audio samples per second, measured in hertz (Hz), typical values: 44,100 Hz or 48,000 Hz
SHA-256	Cryptographic hash algorithm for integrity checking of files
Strict mode	Check mode in which the very first decoding error aborts the check
Stream	An individual data flow within a media file, typically separated into video, audio or subtitles
Subprocess	External program that is started and supervised by VidiVerify, for example a plug-in
Support package	Automatically assembled package of diagnostic information for support
Timeout	Time limit after which an operation is terminated
Transport stream	Continuous data stream with fixed packet sizes, the foundation for digital television
VLC	Plug-in for embedded video playback, also well known as a standalone player
VP9	Video codec, often used inside the WebM container
VQA	Video Quality Assessment, the perceptual quality analysis
Watchdog	Monitoring component that checks the user interface for hung states
WebM	Open container format based on Matroska, usually carrying VP9 or AV1
Yutup	YouTube upload prognosis, a planned PRO module of VidiVerify

Appendix B: Rating system and status classes

VidiVerify assigns every check to a clear status class. This classification is part of the MAP (Media Analysis Protocol) and helps you place the result at a glance. It forms the basis for follow-up decisions such as archiving, further processing or making a new copy.

B.1 Status classes

Status	Meaning
OK	The check completed without errors. Within the chosen check level, the file is considered clean (no issues were detected within the chosen check level).
WARNING	The check completed technically, but produced indications of minor anomalies. A manual review is recommended.
ERROR	The check ran into an error. The file shows structural or decoding problems and should not be used further without inspection.
NOT CHECKED	No result is available yet for this check level.

B.2 Quality classes of the video material

In addition to the technical status class, the MAP assigns the video content to a quality class. The assignment is based on the resolution and offers a quick orientation about the quality of the image material.

Class	Typical resolution	Use case
SD or smaller	up to 720 x 576	Older recordings, mobile recordings from older devices
HD	1280 x 720	Standard web content, streaming, broadcast
Full HD	1920 x 1080	Home cinema, professional productions
2K to 4K	2048 x 1080 to 3840 x 2160	Cinema, high-quality streaming productions
Above 4K	more than 3840 x 2160	Special productions, planetarium, science

B.3 Classification notes

In addition to the technical rating, certain observations are described in text form. Examples of such classifications are:

- Clean structure without anomalies
- Clean check without detectable errors
- Additional streams detected (for example subtitles, chapter markers)
- Timeout during the check

- Faulty file or stream access
- No streams detected
- Plug-in error in FFmpeg, FFprobe or MediaInfo

> **[i] Purpose of the classification**
> The status class gives a quick, color-coded statement about the result. The textual classification adds the cause to this statement, so that even in case of errors it is immediately clear where the problem comes from. Together, these two layers are a core feature of the MAP.

Appendix C: Supported file formats

VidiVerify supports a broad range of common video formats. Detection is based both on the file extension and on the header signature of the file (magic bytes), so that even renamed files are recognized reliably.

C.1 Container formats

Container family	Typical extensions	Description
MP4/MOV (ISO BMFF)	mp4, m4v, m4a, mov, 3gp, 3g2	Widely used, streaming, mobile devices, professional production
Matroska/WebM (EBML)	mkv, webm, mka	Flexible container, often used on the internet
AVI (RIFF)	avi	Classic Windows container format
MPEG program stream	mpg, mpeg, vob, mod, tod	DVD environment, older consumer devices
MPEG transport stream	ts, m2ts, mts	Broadcasting, Blu-ray, IPTV
Windows Media (ASF)	wmv, wma, asf	Older Windows-based productions
Flash Video	flv	Historical web content
MXF	mxf	Broadcast and professional production
Ogg	ogv, ogg	Open media formats
RealMedia	rm, rmvb	Historical streaming content

C.2 Notes on codecs

The actual decoding is handled by the FFmpeg plug-in. All codecs available in the installed version of FFmpeg are supported. These include, among others, H.264, H.265, VP8, VP9, AV1, MPEG-2, MPEG-4, AAC, MP3, FLAC, Opus, PCM and many more.

> **[+] Regular updates recommended**
> New codecs and container variants are continuously integrated into FFmpeg. The automatic update check for the plug-ins makes sure that newer formats can also be recognized and checked reliably.

Appendix D: Access points and controls

VidiVerify is operated entirely through the graphical user interface. Classic keyboard shortcuts such as Ctrl+Shift+P are not currently provided. The following overview summarizes the most important access points and controls.

D.1 Main operation

Action	Access
Choose a file	"Browse" button or drag the file into the application area
Start a check	"Start" button in the main window
Cancel a check	"Cancel" button in the main window
Save the MAP protocol	"Save protocol" button after the check has finished
Start the Diagnostic tool	"Start diagnostics?" button after a standard check with an anomaly, or access it from the Overview tab
"Protocol" tab	Tab in the main window with detailed runtime information
Open Settings	"Settings" button
Create a support package	Entry in the Settings dialog or in the Help menu
Quit the application	Close the window; a confirmation prompt appears while a check is running

D.2 Error handling and diagnostics

- When exceptions occur, an entry is created automatically in exceptions.log
- When processes hang, a diagnostic snapshot is written to the folder logs/hangs
- The device identifier (Machine ID) is displayed in the About dialog and can be shared with support when requesting help
- The support package automatically gathers all relevant diagnostic information

[*] In closing

VidiVerify is designed as a reliable companion for all tasks around quality control of video files. By combining a coordinated set of proven, specialized plug-ins with the proprietary MAP, a clear user interface and open data storage, VidiVerify becomes a tool that fits seamlessly into existing workflows and — together with the planned extensions — is growing into a comprehensive platform for media quality assurance.